CHAPTER 14

Random variables, concentration and the law of large numbers

S62, S64, S73, S74, S77

Key point · Objectives. Calculate the expected value and variance of a sum and of a mean; apply the Bienaymé–Chebyshev inequality; determine a sample size.

Course notes

Sums and transformations

For a finite random variable X, taking the values x_i with probabilities p_i,

$$E(X) = \sum_i x_i p_i, \qquad V(X) = \sum_i (x_i - E(X))^2 p_i = E(X^2) - E(X)^2.$$

The standard deviation is $\sigma(X) = \sqrt{V(X)}$. Expanding the square in $V(aX+b)$, the shift b cancels and the factor a is squared. For finite random variables X, Y and real numbers a, b,

$$E(X+Y) = E(X) + E(Y), \quad E(aX+b) = aE(X) + b, \quad V(aX+b) = a^2 V(X).$$

The linearity of the expected value does not require independence. If X, Y are independent, $V(X+Y) = V(X) + V(Y)$ (accepted without proof). Without this assumption, this formula cannot be applied. The standard deviation of $aX+b$ is $|a|\sigma(X)$.

Example of dependence. If $Y = X$, then $V(X+Y) = V(2X) = 4V(X)$, and not $2V(X)$ as soon as $V(X) \neq 0$.

Proof for the binomial distribution. Write $X = B_1 + \cdots + B_n$ with the independent indicators of the successes. Each B_i satisfies $E(B_i) = p$, $E(B_i^2) = p$ and $V(B_i) = p - p^2$. By linearity and additivity under independence, $E(X) = np$, $V(X) = np(1-p)$.

Sample and mean

A sample of size n of a distribution is a family of n independent random variables with the same distribution. If they have expected value μ and variance σ^2, we write $S_n = \sum_{i=1}^n X_i$ and $M_n = \frac{S_n}{n}$.

Key point · Sum and mean.

$$E(S_n) = n\mu, \quad V(S_n) = n\sigma^2, \qquad E(M_n) = \mu, \quad V(M_n) = \frac{\sigma^2}{n}, \quad \sigma(M_n) = \frac{\sigma}{\sqrt{n}}.$$

To divide the standard deviation of the mean by 2, multiply the size by 4. The variance of the sum increases; that of the mean decreases.

Concentration

Key point · Bienaymé–Chebyshev inequality. For a random variable with expected value μ, variance σ^2, and $\delta > 0$,

$$P(|X - \mu| \geq \delta) \leq \frac{\sigma^2}{\delta^2}.$$

An upper bound greater than 1 remains true, but gives no further information; it can be replaced by 1.

Proof in the finite case. In $V(X) = \sum_x (x - \mu)^2 P(X = x)$, keep only the values satisfying $|x - \mu| \geq \delta$. Each square is then at least δ^2. Hence $V(X) \geq \delta^2 P(|X - \mu| \geq \delta)$.

Law of large numbers

Applied to the mean of a sample,

$$P(|M_n - \mu| \geq \delta) \leq \frac{\sigma^2}{n\delta^2} \xrightarrow[n\to\infty]{} 0.$$

This is the law of large numbers in the setting considered: for a fixed precision, the probability of a deviation at least equal to this precision tends to zero. This result does not say that each realisation approaches μ monotonically.

Method · Choosing a sufficient size. To guarantee $P(|M_n - \mu| < \delta) \geq 1 - \alpha$, it suffices to choose

$$n \geq \frac{\sigma^2}{\alpha\delta^2}.$$

Take the ceiling. This gives the smallest integer satisfying *this bound*, not necessarily the minimum size for the exact probability.

For a frequency of successes, $\mu = p$ and $\sigma^2 = p(1 - p) \leq \frac{1}{4}$. If p is unknown, $n \geq \frac{1}{4\alpha\delta^2}$ is a uniform guarantee.

Research activity

Fluctuations of a frequency — 40 min.

A fair coin is tossed independently. For every integer $n \geq 1$, let F_n be the frequency of "heads" among the first n tosses.

1. Write a program that simulates 1000 tosses, then plot the points (n, F_n) for $1 \leq n \leq 1000$. Does the deviation from $\frac{1}{2}$ necessarily decrease at every toss?
2. Express $E(F_n)$ and $V(F_n)$ in terms of n.
3. Using Bienaymé–Chebyshev, find an upper bound for $P(|F_{1000} - \frac{1}{2}| \geq 0.05)$.
4. Determine an integer n sufficient for $P(|F_n - \frac{1}{2}| < 0.05) \geq 0.99$.
5. State what the simulated trajectory shows and what the inequality obtained guarantees. Does a single simulation constitute a proof of this guarantee?

Progressive exercises

E1 · Practice. Let X be a random variable such that $E(X) = 3$ and $V(X) = 2$. Let $Y = 4X - 1$. Calculate the expected value, the variance and the standard deviation of Y.

E2 · Practice. The random variables X and Y are independent and satisfy $E(X) = 2$, $E(Y) = 5$, $V(X) = 1$ and $V(Y) = 4$. Calculate $E(2X - Y)$ and $V(2X - Y)$.

E3 · Practice. Consider 64 independent random variables with the same distribution, with expected value 10 and standard deviation 4. Calculate the expected value and the standard deviation of their mean.

E4 · Practice. Let X be a random variable such that $E(X) = 100$ and $V(X) = 9$. Using the Bienaymé–Chebyshev inequality, give an upper bound for $P(|X - 100| \geq 6)$.

E5 · Practice. Let M_n be the mean of n independent random variables with the same distribution, with expected value μ and variance 4. Determine a size n sufficient for $P(|M_n - \mu| < 0.2) \geq 0.95$.

Textbook or past-paper work

Baccalauréat, Metropolitan France, 17 June 2025: exercise 1, question 6 (p. 3). At home after S77. Read the context of question 5 and use the expected value and variance values accepted in question 6. The paper is available here: official past paper. Use the data and instructions of this past paper.

Homework

Homework 14 — Quality of an estimate — 55 min.

We repeat $n \geq 1$ independent Bernoulli trials with the same parameter $p \in [0,1]$, which is unknown. Let F_n be the frequency of successes.

1. Prove that $E(F_n) = p$ and $V(F_n) = \frac{p(1-p)}{n}$.
2. Show that $p(1-p) \leq \frac{1}{4}$ for all $p \in [0,1]$.
3. Deduce a size n sufficient for $P(|F_n - p| < 0.02) \geq 0.95$, whatever the value of p.
4. We now know that $p \leq 0.1$. Determine a sharper upper bound for $p(1-p)$, then improve the sufficient size obtained in question 3.
5. For this question only, the n results are all equal to the same Bernoulli variable B with parameter p. Calculate F_n and its variance. Explain why the previous guarantee no longer applies.

Teaching notes and solutions

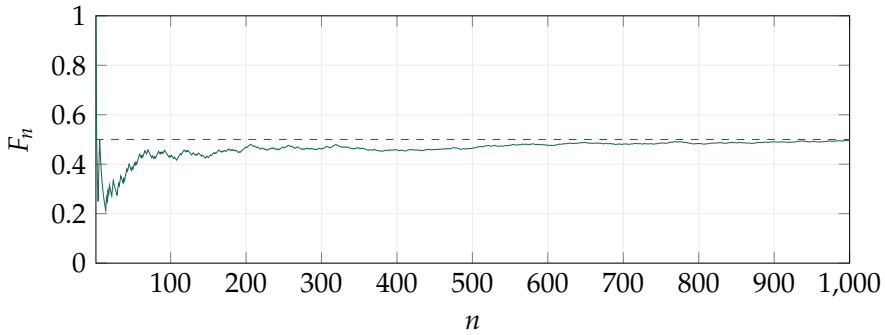
Teacher note · Teacher notes. Test 5 and the second mock exam come before S73: do not ask for Bienaymé–Chebyshev or the law of large numbers in them. For the variance, have students circle the independence assumption. For concentration, distinguish the exact probability from the bound, and a sufficient rank from the true minimal rank. A simulation is not a proof, and an observed mean does not necessarily get closer with each new trial.

Solutions to the activity

1.
```python
from random import Random

def trajectoire_piece(N=1000, graine=2026):
    rng = Random(graine)
    piles, frequences = 0, []
    for n in range(1, N+1):
        piles += rng.random() < 0.5
        frequences.append(piles/n)
    return frequences
```

Plot the points (n, F_n) with the list obtained. After two tosses giving one head and one tail, $F_2 = \frac{1}{2}$. Whatever the third result, F_3 is $\frac{1}{3}$ or $\frac{2}{3}$: the deviation from $\frac{1}{2}$ increases. Stabilisation is not a monotonic decrease of the deviation along each trajectory.



A simulated trajectory, seed 2026; the line is $y = \frac{1}{2}$.

2. Let B_i be the indicator of heads. Then $F_n = \frac{B_1+\cdots+B_n}{n}$. Each B_i has expected value $\frac{1}{2}$ and variance $\frac{1}{4}$. By linearity, $E(F_n) = \frac{1}{2}$; by independence and rescaling, $V(F_n) = \frac{n(\frac{1}{4})}{n^2} = \frac{1}{4n}$.
3. Bienaymé–Chebyshev gives $P(|F_{1000} - \frac{1}{2}| \geq 0.05) \leq \frac{1}{4\times 1000\times 0.05^2} = 0.1$. This is an upper bound, not the exact probability.
4. To ensure a complementary probability of at most 0.01, it suffices that $\frac{1}{4n\times 0.05^2} \leq 0.01$. This is equivalent to $n \geq 10000$. Then $P(|F_n - \frac{1}{2}| < 0.05) \geq 0.99$.
5. The simulation illustrates one finite realisation; other seeds give other trajectories. The inequality establishes a guarantee on the probability, for any sample satisfying the assumptions of independence and common distribution.

E1. The affine transformation gives $E(4X-1) = 4E(X) - 1 = 11$ and $V(4X-1) = 4^2V(X) = 32$. The standard deviation is $\sqrt{32} = 4\sqrt{2}$; the shift -1 does not change the dispersion.

E2. By linearity, $E(2X-Y) = 2E(X) - E(Y) = 4 - 5 = -1$. The independence of X, Y allows us to add the variances of $2X$ and $-Y$: $V(2X-Y) = 4V(X) + V(Y) = 4 + 4 = 8$.

E3. The mean of 64 independent variables with the same distribution keeps the expected value 10. Its variance is $\frac{4^2}{64} = \frac{1}{4}$, so its standard deviation is $\frac{1}{2}$.

E4. Bienaymé–Chebyshev with $\delta = 6$ gives $P(|X - 100| \geq 6) \leq \frac{V(X)}{6^2} = \frac{9}{36} = \frac{1}{4}$. This bound gives neither an equality nor an exact value.

E5. We have $P(|M_n - \mu| \geq 0.2) \leq \frac{4}{n \times 0.2^2} = \frac{100}{n}$. For this bound to be at most 0.05, it suffices that $n \geq 2000$. The complement then has probability at least 0.95.

Solutions to Homework 14

1. Let B_i be the independent indicators of the successes. $F_n = (\frac{1}{n}) \sum_{i=1}^{n} B_i$, with $E(B_i) = p$, $V(B_i) = p(1-p)$. Hence
$$E(F_n) = \frac{1}{n} \sum_{i=1}^{n} p = p, \qquad V(F_n) = \frac{1}{n^2} \sum_{i=1}^{n} p(1-p) = \frac{p(1-p)}{n}.$$
2. $p(1-p) = p - p^2 = \frac{1}{4} - (p - \frac{1}{2})^2 \leq \frac{1}{4}$. Equality is attained for $p = \frac{1}{2}$.
3. For all $p \in [0,1]$, $P(|F_n - p| \geq 0.02) \leq \frac{1}{4n \times 0.02^2}$. It suffices that this bound be at most 0.05, that is $n \geq 12500$. By complement, this size guarantees the required probability, uniformly in p.
4. On $[0, 0.1]$, the function $p \mapsto p(1-p)$ is increasing because its derivative $1 - 2p$ is positive. Hence $p(1-p) \leq 0.1 \times 0.9 = 0.09$. It now suffices that $n \geq \frac{0.09}{0.05 \times 0.02^2} = 4500$. These sizes are sufficient in the sense of the bound, not necessarily minimal for the exact probability.
5. If all the trials copy the same variable B, then $F_n = \frac{B + \cdots + B}{n} = B$. Hence $V(F_n) = p(1-p)$, with no division by n. Adding the variances was allowed only because of independence; this assumption disappears here. For $0 < p < 1$, increasing the number of copies brings no new information.

Solutions to the textbook or past-paper work

M_N is the mean number of universal donors per sample of 100 people. Linearity gives $E(M_N) = 7.14$; independence gives $V(M_N) = \frac{6.63}{N}$.

$$P(|M_N - 7.14| \geq 0.14) \leq \frac{6.63}{0.14^2 N}.$$

To make this bound at most 0.05, it suffices to take

$$N \geq \left\lceil \frac{6.63}{0.05 \times 0.14^2} \right\rceil = 6766.$$

By complement, $P(7 < M_N < 7.28) \geq 0.95$.